

need for fuel replenishment, storage tanks, and distribution infrastructure. The device requires only 100 millilitres of water and approximately 1kWh electricity to operate.

From water to flame: The science at the core

The operating principle of a hydrogen stove is straightforward. It relies on electrolysis, in which electricity splits water into hydrogen and oxygen. This reaction occurs in a PEM electrolyser, where hydrogen is generated at the cathode and oxygen is released at the anode.

Unlike older systems that depended on high-pressure hydrogen storage, these stoves generate hydrogen only when required. As a result, dedicated fuel storage and transportation systems may no longer be necessary.

The PEM electrolyzers used in small-scale systems typically operate at 1.8V to 2.2V and current densities of 1A/cm² to 2A/cm². These parameters influence hydrogen production rates, temperature management, and system longevity. Water quality, temperature regulation, and electrode degradation are also critical factors in these small-scale systems.

Electricity, rather than water, is the true source of energy. Electrolysis typically operates at 65-75% efficiency, while additional losses occur during hydrogen combustion, making overall efficiency broadly comparable to LPG but lower than that of induction cookers.

During combustion, hydrogen is converted primarily into water vapour, producing very little direct air pollution. At first glance, this makes it appear to be an exceptionally clean fuel. However, the key consideration is the source of the electricity used to generate the hydrogen.

Safety presents another significant challenge. Hydrogen ignites easily, and its flame is nearly invisible to the naked eye. In practical situations, this can make leaks

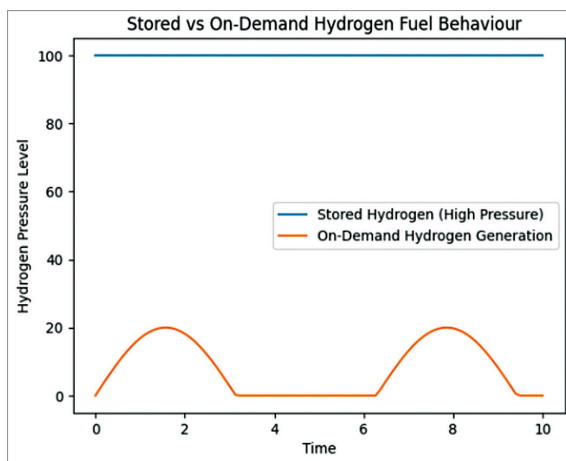


Fig. 1: Hydrogen in this system is not a stored resource—it is a real-time event

or fires difficult to detect quickly. This characteristic can significantly increase operational risk in real-world applications.

To address these challenges, modern hydrogen cooking systems are carefully engineered. They typically incorporate durable, heat-resistant burner materials such as ceramics and silicon carbide (SiC) to enhance operational safety. In addition, catalysts such as platinum are used to improve reaction efficiency and provide better process control.

This raises an important question: Is hydrogen truly a clean fuel, or is it only as clean as the energy used to produce it?

Inside the appliance: A micro energy system

At a systems level, a hydrogen stove can be viewed as a compact energy-conversion device. AC power from the grid is converted to DC and regulated via DC-to-DC conversion before being supplied to the PEM electrolysis cell. There, hydrogen and oxygen are generated, separated, and controlled through microvalves before being delivered to a specialised burner.

The design also incorporates water input systems (using distilled or RO water), power electronics, the PEM electrolyser, gas-management systems, and a sensor-controlled management unit. Sensors connected to microcontrollers enable control

techniques such as PWM, ensuring precise hydrogen production. By integrating IoT and AI, the system can optimise energy use and adapt to changing electricity prices.

From the user's perspective, the experience remains largely unchanged: instant heat, a visible flame, and compatibility with cooking methods such as roasting and grilling—capabilities not easily

replicated by conventional induction cookers. The stove generates minimal direct emissions during operation and offers efficiency levels comparable to LPG, although slightly lower than induction cooking. Requiring only water and electricity as inputs, it can operate as a renewable cooking solution when powered by solar energy.

In simple terms, the device functions as a distributed, real-time energy system that converts electrical energy into cooking fuel at the point of use.

The Indian context: Opportunity meets constraint

Although hydrogen offers significant potential as a cooking fuel, major challenges remain around cost, policy direction, and technology deployment in India. Through the Pradhan Mantri Ujjwala Yojana, many households can now afford LPG connections, resulting in improved health outcomes and living standards. However, because refilling cylinders remains expensive, many families use LPG only for supplementary cooking needs and continue relying on traditional biomass fuels for the remainder.

The National Green Hydrogen Mission is helping accelerate interest in hydrogen technologies, particularly in the industrial and transport sectors, while decentralised applications such as cooking continue to progress more gradually. Advances in miniaturised